

# Statically determinate / Statically indeterminate

## Statically determinate

### Stable

- All unknown reactive forces can be determined

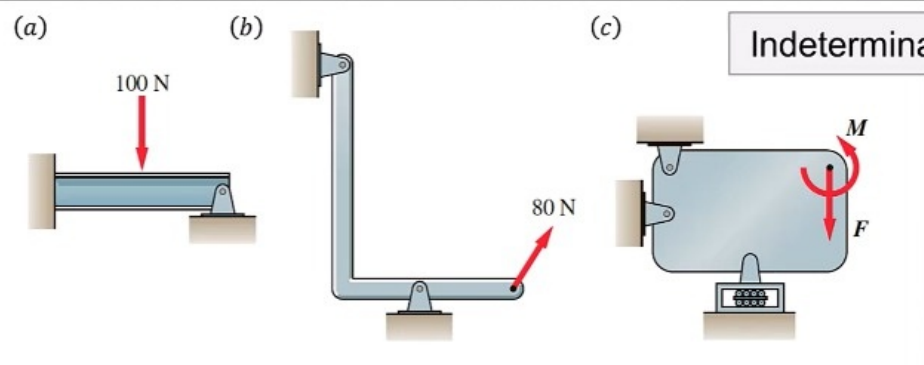
[System of eqn is determined/over-determined]

## Statically indeterminate

### Stable

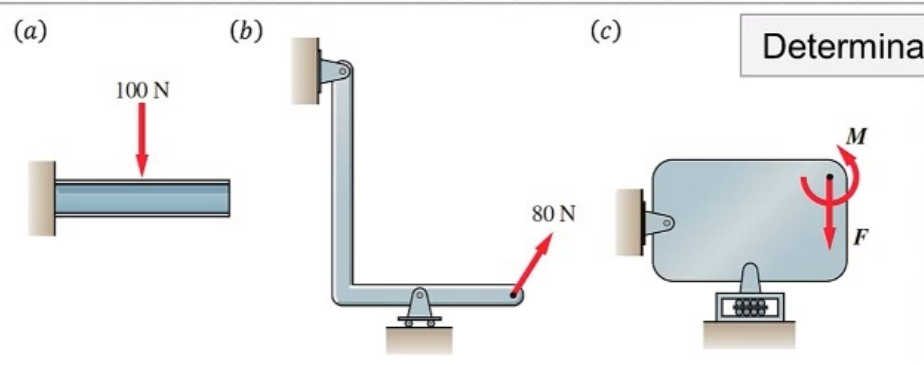
- All unknown reactive forces can't be determined

[System of eqn is underdetermined]



Indeterminate systems

- (a) 5 unknowns, 3 equations
- (b) 4 unknowns, 3 equations
- (c) 5 unknowns, 3 equations



Determinate systems

- (a) 3 unknowns, 3 equations
- (b) 3 unknowns, 3 equations
- (c) 3 unknowns, 3 equations

## Improper constraints

- ~ Smtg isn't properly held in place
  - ↳ Move too much when it shouldn't
  - ↳ Held too tight and might break

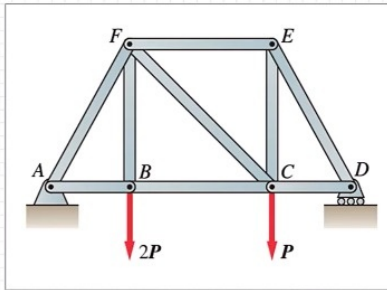
# Trusses

~ Strong framework made w straight pieces [Members] connected tgl to form triangles

## Truss determinacy

The truss' determinacy can be evaluated by counting the number of two-force members ( $m$ ), the number of joints ( $j$ ), and the number of reactions ( $r$ ).

- A **statically determinate** truss obeys  $m + r = 2j$
- A **statically indeterminate** truss will follow  $m + r > 2j$  (excess members or reactions)

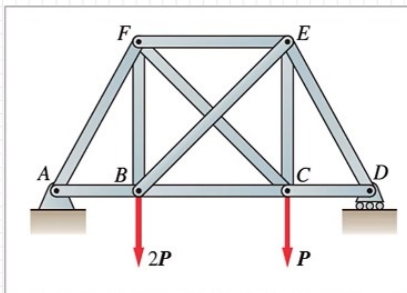


$m = 9$  members  
 $r = 3$  reactions  
 $j = 6$  joints

$m + r = 12$   
 $2j = 12$   
 $\therefore m + r = 2j$   
Truss is determinate

The truss' determinacy can be evaluated by counting the number of two-force members ( $m$ ), the number of joints ( $j$ ), and the number of reactions ( $r$ ).

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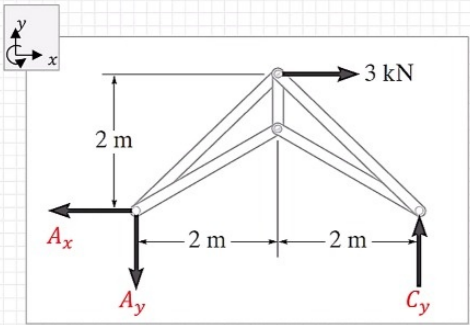
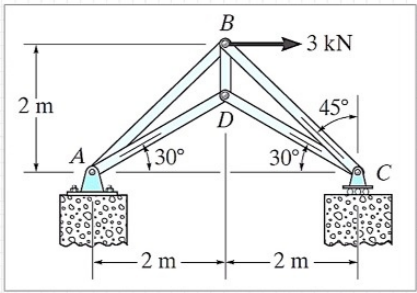


$m = 10$  members  
 $r = 3$  reactions  
 $j = 6$  joints

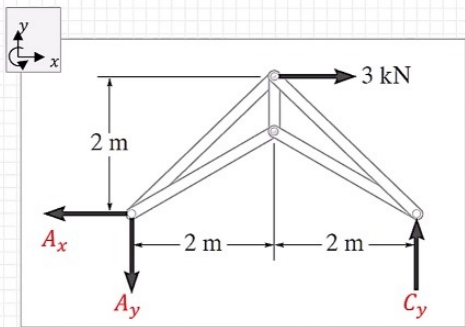
$m + r = 13$   
 $2j = 12$   
 $\therefore m + r > 2j$   
Truss is indeterminate

## Example 1

The reactions of a truss can be calculated by drawing a free-body diagram of the truss and replacing the supports with its reactions. In this example, there is a pin support at point *A* and a roller support at point *C*.



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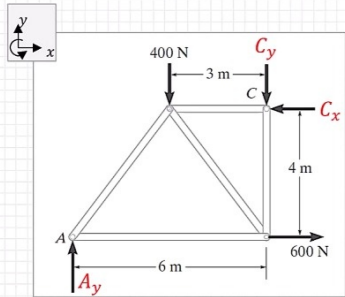
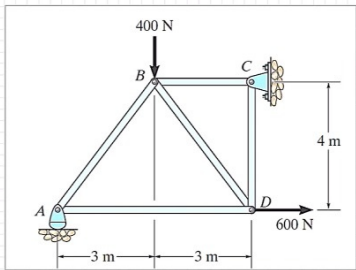
$$\begin{aligned}\sum F_x &= 0 \\ -A_x + 3 &= 0 \\ A_x &= 3 \text{ kN}\end{aligned}$$

$$\begin{aligned}\sum M_A &= 0 \\ C_y(4) - 3(2) &= 0 \\ C_y &= 1.5 \text{ kN}\end{aligned}$$

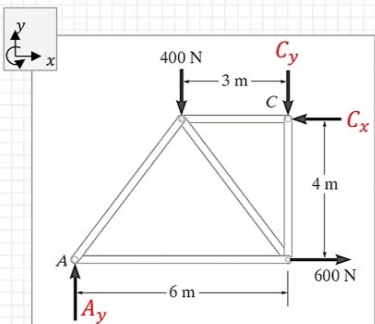
$$\begin{aligned}\sum F_y &= 0 \\ -A_y + C_y &= 0 \\ A_y &= 1.5 \text{ kN}\end{aligned}$$

## Example 2 :

Determine the reactions at points *A* and *C*.



Determine the reactions at points *A* and *C*.



$$\begin{aligned}\sum F_x &= 0 \\ -C_x + 600 &= 0 \\ C_x &= 600 \text{ N}\end{aligned}$$

$$\begin{aligned}\sum M_C &= 0 \\ -A_y(6) + 400(3) + 600(4) &= 0 \\ A_y &= 600 \text{ N}\end{aligned}$$

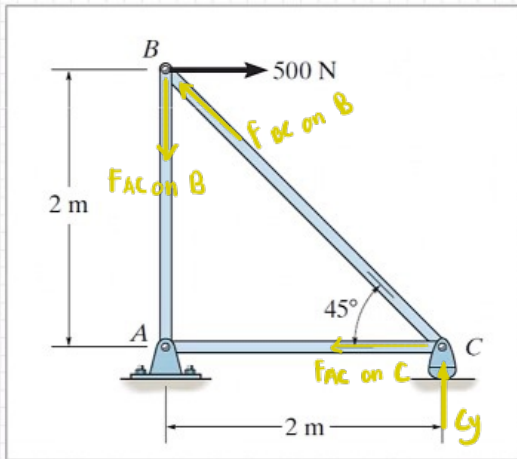
$$\begin{aligned}\sum F_y &= 0 \\ A_y - 400 - C_y &= 0 \\ C_y &= 200 \text{ N}\end{aligned}$$

## Method of joints

~ Finding member force by applying joint equilibrium to every joint in truss

\* One joint only has 2 eqn ( $\sum F_x$  &  $\sum F_y$ ), thus solve joints w/  $\leq 2$  unknowns 1<sup>st</sup>.

Determine the forces in each member of the truss using the method of joints and indicate if the members are in tension or compression.



### Joint B [Forces ON B]

$$\sum F_x = 0$$

$$500 - F_{BC} \sin(45^\circ) = 0$$

$$F_{BC} = 707.1 \text{ N}$$

$\therefore$  Since the assumed direc. is correct.

• BC pushes on B

$\rightarrow$  B pushes on BC

$\rightarrow$  BC gets compressed.

$$\sum F_y = 0$$

$$F_{BC} \cos(45^\circ) - F_{BA} = 0$$

$$F_{BA} = 500 \text{ N}$$

$\therefore$  Since the assumed direc. is correct.

• BA pulls on B

$\rightarrow$  B pulls on BC

$\rightarrow$  BC gets stretched (tension)

### Joint C [Forces ON C]

$$\sum F_x = 0$$

$$F_{BC} \cos(45^\circ) - F_{CA} = 0$$

$$F_{CA} = 500 \text{ N}$$

$\therefore$  Since the assumed direc. is correct.

• CA pulls on C

$\rightarrow$  C pulls on CA

$\rightarrow$  CA gets stretched (tension)

\* Member in Tension

$\rightarrow$  Force by member away from joint

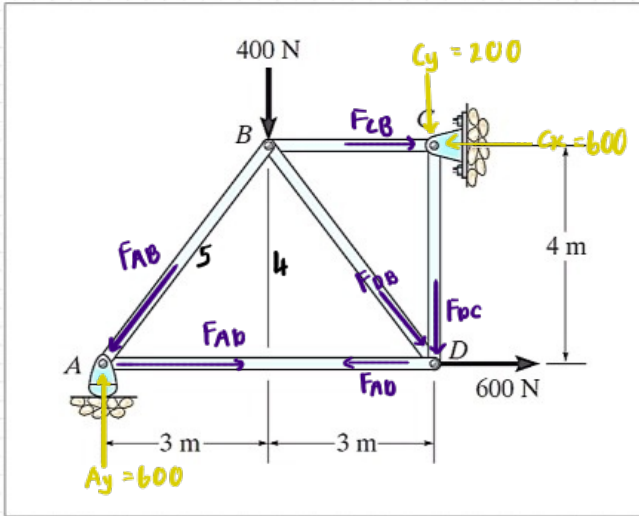
Member in Compression

$\rightarrow$  Force by member towards joint



## Example

Determine the forces in each member of the truss using the method of joints and indicate if the members are in tension or compression.



$$\begin{aligned}\sum F_x &= 0 \\ 600 - C_x &= 0 \\ C_x &= 600 \text{ N}\end{aligned}$$

$$\begin{aligned}\sum \mathcal{L}_C &= 0 \\ 400(3) + 600(4) - 6A_y &= 0 \\ A_y &= 600 \text{ N}\end{aligned}$$

$$\begin{aligned}\sum F_y &= 0 \\ A_y - 400 - C_y &= 0 \\ C_y &= 200 \text{ N}\end{aligned}$$

Beac 1 joint only have  
2 eqn ( $\sum F_x$  &  $\sum F_y$ )

Choose joints w  $\leq 2$  unknowns  
to solve 1st

Force on Joint A

$$\begin{aligned}\sum F_y &= 0 \\ 600 - F_{AB}(\frac{4}{5}) &= 0 \\ F_{AB} &= 750 \text{ N} \# \end{aligned}$$

$\therefore$  AB pushes A, A pushes AB  
 $\hookrightarrow$  AB is in compression

$$\begin{aligned}\sum F_x &= 0 \\ F_{AD} - F_{AB}(\frac{3}{5}) &= 0 \\ F_{AD} &= 450 \text{ N} \# \end{aligned}$$

$\therefore$  AD pulls on A, A pulls on AD  
 $\hookrightarrow$  AD is in tension

Force on joint D

$$\begin{aligned}\sum F_x &= 0 \\ 600 + F_{DB}(\frac{3}{5}) - 450 &= 0 \\ F_{DB} &= -250 \text{ N} \# \end{aligned}$$

$\therefore$  Since -ve, direc. is opposite  
of assumed. DB pulls on D,  
D pulls on DB  
 $\hookrightarrow$  DB is in tension

$$\begin{aligned}\sum F_y &= 0 \\ -F_{DB}(\frac{4}{5}) - F_{DC} &= 0 \\ F_{DC} &= -(-250)(\frac{4}{5}) \\ &= 200 \text{ N} \# \end{aligned}$$

$\therefore$  DC pushes D, D pushes  
 $\hookrightarrow$  DC is in compression

Force on joint C

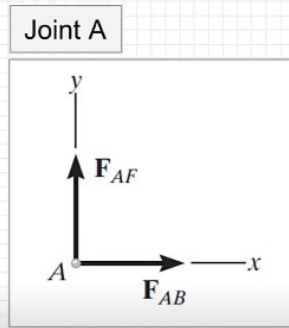
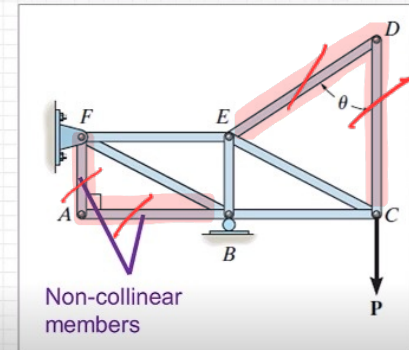
$$\begin{aligned}\sum F_x &= 0 \\ F_{CB} - 600 &= 0 \\ F_{CB} &= 600 \text{ N} \# \end{aligned}$$

$\therefore$  CB pushes C, C pushes  
 $\hookrightarrow$  CB is in compression

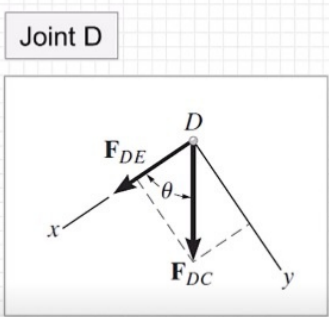
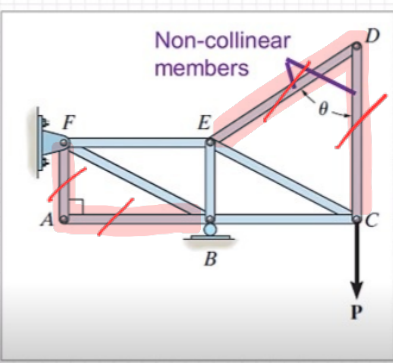
## Zero force members

~ Members that do not carry any load

Rule 1: If only two non-collinear members form a truss joint and no external load or support reaction is applied to the joint, then the two members must be zero-force members.

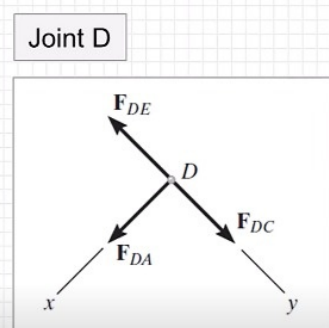
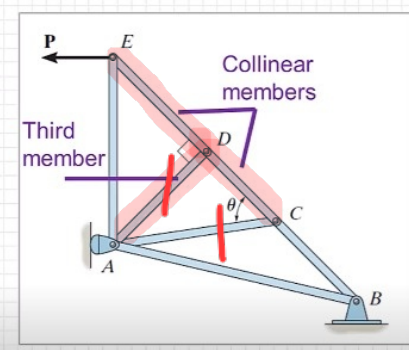


$$\begin{aligned}\sum F_x &= 0 \\ F_{AB} &= 0 \text{ N} \\ \sum F_y &= 0 \\ F_{AF} &= 0 \text{ N}\end{aligned}$$

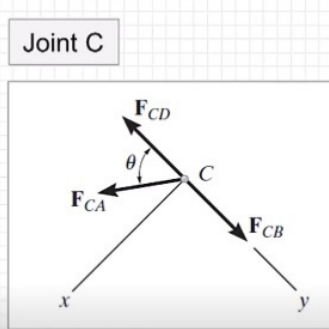
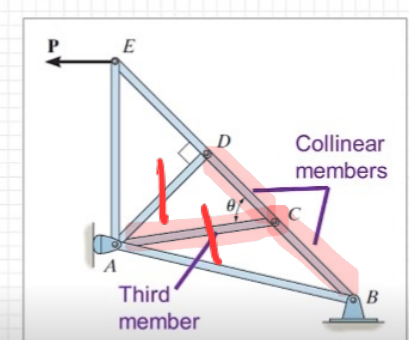


$$\begin{aligned}\sum F_y &= 0 \\ F_{DC} \sin(\theta) &= 0 \\ F_{DC} &= 0 \text{ N since } \sin(\theta) \neq 0 \text{ for } \theta > 0 \\ \sum F_x &= 0 \\ F_{DE} + F_{DC} \cos(\theta) &= 0 \\ F_{DE} + 0 \cos(\theta) &= 0 \\ F_{DE} &= 0 \text{ N}\end{aligned}$$

Rule 2: If three members form a truss joint for which two of the members are collinear, then the third member is a zero-force member provided there is no external force or support reaction that has a component that acts along the third member.



$$\begin{aligned}\sum F_x &= 0 \\ F_{DA} &= 0 \text{ N} \\ \sum F_y &= 0 \\ -F_{DE} + F_{DC} &= 0 \\ F_{DE} &= F_{DC}\end{aligned}$$



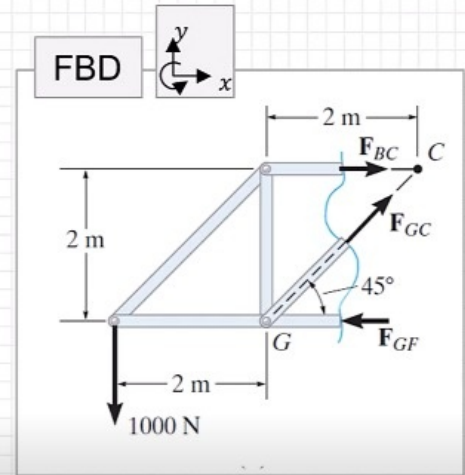
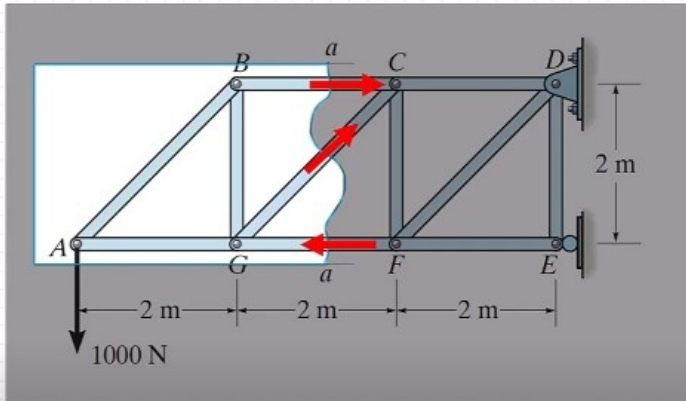
$$\begin{aligned}\sum F_x &= 0 \\ F_{CA} \sin(\theta) &= 0 \\ F_{CA} &= 0 \text{ N since } \sin(\theta) \neq 0 \text{ for } \theta > 0 \\ \sum F_y &= 0 \\ -F_{CD} - F_{CA} \cos(\theta) + F_{CB} &= 0 \\ F_{CD} - 0 + F_{CB} &= 0 \\ F_{CD} &= F_{CB}\end{aligned}$$

## Method of sections

~ Cut the truss into sections and apply  $\sum F_x = 0$ ,  $\sum F_y = 0$ .

[Beuz a system in eq. means any subset will also be in eq.]

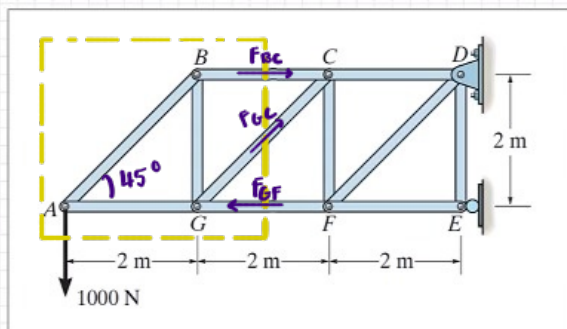
Generally, our free-body diagram does not  <sup>$\sum F_x, \sum F_y, \sum F_z$</sup>  cut through more than 3 members (i.e. 3 unknowns) since we only have 3 equilibrium equations. We should be able to apply equilibrium equations to solve for the unknown forces.





### Example !

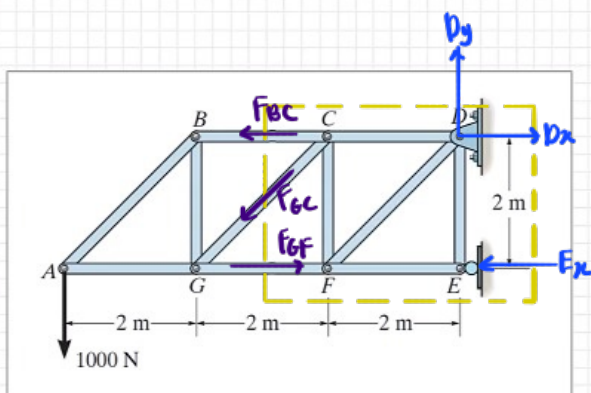
Determine the forces in members  $BC$ ,  $CG$ , and  $GF$  using the method of sections and indicate if the members are in tension or compression.



|                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\sum \mathcal{L}_C = 0$ $1000(4) - F_{GF}(2) = 0$ $F_{GF} = 1000 \text{ N} \#$ <p style="text-align: center;">[compression]</p> <hr/> $\sum \mathcal{L}_G = 0$ $1000(2) - F_{BC}(2) = 0$ $F_{BC} = 1000 \text{ N} \#$ <p style="text-align: center;">[tension]</p> | $\sum F_y = 0$ $F_{GC} \sin(45^\circ) - 1000 = 0$ $F_{GC} = 1414.214 \text{ N} \#$ <p style="text-align: center;">[tension]</p> <hr/> $\sum F_x = 0$ $F_{BC} + F_{GC} \cos(45^\circ) - F_{GF} = 0$ $F_{GC} = 1414.214 \text{ N} \#$ <p style="text-align: center;">[check]</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Alternate method

Determine the forces in members  $BC$ ,  $CG$ , and  $GF$  using the method of sections and indicate if the members are in tension or compression.



[Use entire diagram]  
① Find support rxn

$$\sum F_y = 0$$

$$D_y - 1000 = 0$$

$$D_y = 1000$$


---


$$\sum \mathcal{L}_B = 0$$

$$1000(6) - E_x(2) = 0$$

$$E_x = 3000 \text{ N}$$


---


$$\sum F_x = 0$$

$$D_x - E_x = 0$$

$$D_x = 3000 \text{ N}$$

② Solve for members

$$\sum \mathcal{L}_C = 0$$

$$F_{GF}(2) - E_x(2) + D_y(2) = 0$$

$$F_{GF} = 2000 \text{ N} \#$$

[compression]

---


$$\sum \mathcal{L}_G = 0$$

$$F_{BC}(2) + D_y(4) - D_x(2) = 0$$

$$F_{BC} = 1000 \text{ N} \#$$

[tension]

---


$$\sum F_y = 0$$

$$D_y - F_{GC} \sin(45^\circ) = 0$$

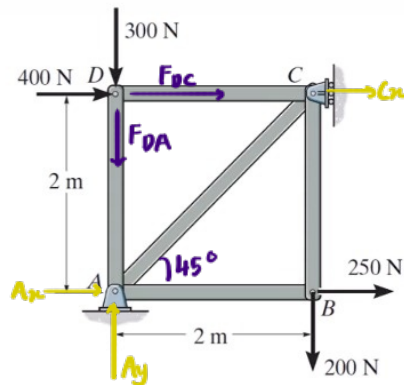
$$F_{GC} = 1414.214 \text{ N} \#$$

[tension]



## Extra Ques

- 1a) Consider the truss below which is supported on a pin joint at point A and a roller joint at point C. Determine the force in each member using the method of joints. Use a positive value to indicate tension and a negative value to indicate compression.



### ① Find rxn forces

$$\sum F_y = 0$$

$$A_y - 300 - 200 = 0$$

$$A_y = 500 \text{ N}$$

$$\sum \mathcal{M}_A = 0$$

$$-400(2) - C_x(2) - 200(2) = 0$$

$$C_x = -600 \text{ N}$$

$$\sum F_x = 0$$

$$400 + A_x + C_x + 250 = 0$$

$$A_x = -50 \text{ N}$$

$$\sum F_y = 0$$

$$-F_{DA} - 300 = 0$$

$$F_{DA} = -300 \text{ N [Compression]}$$

### On Joint C

$$\sum F_x = 0$$

$$-F_{CB} - F_{CA} \cos 45^\circ + C_x = 0$$

$$F_{CA} = -282.843 \text{ N [Compression]}$$

$$\sum F_y = 0$$

$$-F_{CA} \sin 45^\circ - F_{CB} = 0$$

$$F_{CB} = 200 \text{ N [Tension]}$$

### ② Solve for members

#### On Joint D

$$\sum F_x = 0$$

$$F_{DC} + 400 = 0$$

$$F_{DC} = -400 \text{ N [Compression]}$$

#### On Joint B

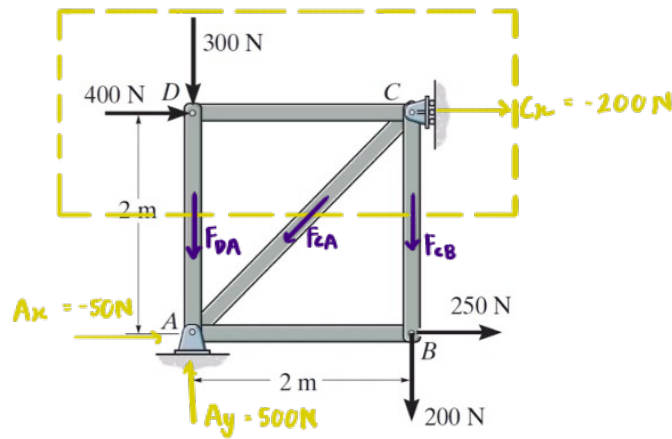
$$\sum F_x = 0$$

$$-F_{BA} + 250 = 0$$

$$F_{BA} = 250 \text{ N [Tension]}$$

1b)

Consider the truss below which is supported on a pin joint at point A and a roller joint at point C. Determine the force in members AD, AC, and BC using the method of sections. Use a positive value to indicate tension and a negative value to indicate compression.



$$\sum \mathcal{C}_C = 0$$

$$300(2) + F_{DA}(2) = 0$$

$$F_{DA} = -300 \text{ N [Compression]}$$

$$\sum \mathcal{C}_A = 0$$

$$-400(2) - C_x(2) - F_{CB}(2) = 0$$

$$F_{CB} = 200 \text{ N [Tension]}$$

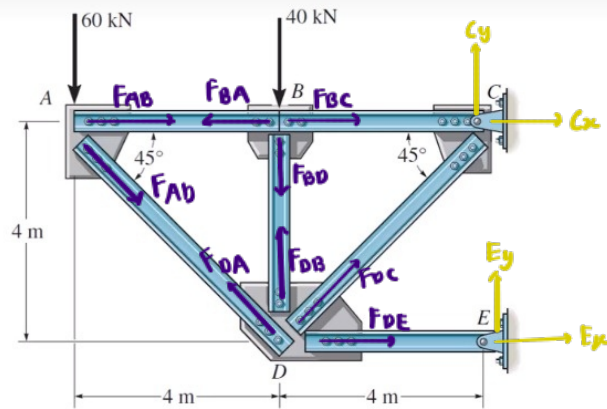
$$\sum F_x = 0$$

$$400 - F_{CA} \cos 45^\circ + C_x = 0$$

$$F_{CA} = -282.843 \text{ N [compression]}$$

2a)

Consider the truss below which is supported on pin joints at points C and E. Determine the force in each member using the method of joints. Use a positive value to indicate tension and a negative value to indicate compression.



### ① Solve rxn forces

$$\sum \mathcal{C}_x = 0$$

$$60(8) + 40(4) + E_x(4) = 0$$

$$E_x = -160 \text{ N}$$

$$\sum F_x = 0$$

$$C_x + E_x = 0$$

$$C_x = 160 \text{ N}$$

$$\sum F_y = 0$$

$$-60 - 40 + C_y + E_y = 0$$

### On Joint B

$$\sum F_y = 0$$

$$-40 - F_{BD} = 0$$

$$F_{BD} = -40 \text{ kN [Compression]} \#$$

$$\sum F_x = 0$$

$$-F_{BA} + F_{BC} = 0$$

$$F_{BC} = 60 \text{ kN [Tension]} \#$$

### On Joint D

$$\sum F_y = 0$$

$$F_{DA} \cos 45^\circ + F_{DB} + F_{DC} \cos 45^\circ = 0$$

$$F_{DC} = 141.4214 \text{ kN [Tension]} \#$$

### On Joint A

$$\sum F_y = 0$$

$$-60 - F_{AD} \sin 45^\circ = 0$$

$$F_{AD} = -84.85 \text{ kN [Compression]} \#$$

$$\sum F_x = 0$$

$$F_{AB} + F_{AD} \cos 45^\circ = 0$$

$$F_{AB} = 60 \text{ kN [Tension]} \#$$

$$\sum F_x = 0$$

$$-F_{DA} \sin 45^\circ + F_{DC} \sin 45^\circ + F_{DE} = 0$$

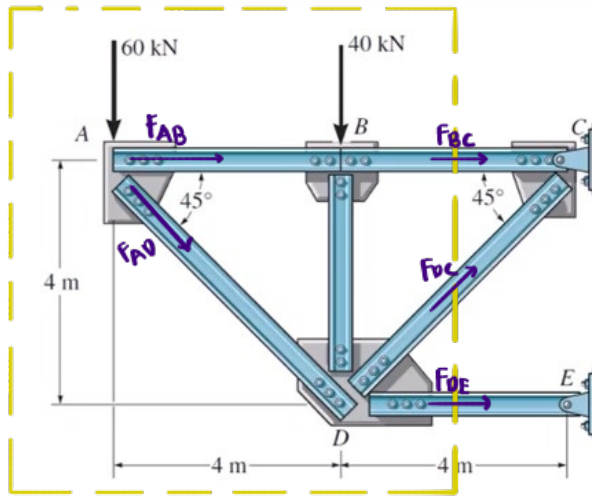
$$-(-84.8528) \sin 45^\circ + 141.4214 \sin 45^\circ + F_{DE} = 0$$

$$F_{DE} = -160 \text{ kN [Compression]} \#$$



2b)

Consider the truss below which is supported on pin joints at points C and E. Determine the force in members BC, DC, and DE using the method of sections. Use a positive value to indicate tension and a negative value to indicate compression.



$$\sum \mathcal{L}_D = 0$$

$$60(4) - F_{BC}(4) = 0$$

$$F_{BC} = 60 \text{ kN [In Tension]} *$$

$$\sum \mathcal{L}_C = 0$$

$$60(8) + 40(4) + F_{DE}(4) = 0$$

$$F_{DE} = -160 \text{ kN [In Compression]}$$

$$\sum F_y = 0$$

$$-60 - 40 + F_{DC} \sin 45^\circ = 0$$

$$F_{DC} = 141.4214 \text{ kN [In Tension]}$$